IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2024

MSc and EEE/EIE PART IV: MEng and ACGI

OPTICAL COMMUNICATION

Wednesday, 15 May 10:00 am

Time allowed: 3:00 hours

There are FIVE questions on this paper.

Answer Question ONE, and ANY THREE of Questions 2 to 5

All questions carry equal marks.

Any special instructions for invigilators and information for candidates are on page 1.

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Second Marker(s) : S. Lucyszyn

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Special instructions for invigilators: None.

Information for Candidates:

Numbers in brackets in the right margin (e.g. [5]) indicate maximum marks for each section of each question.

The following constants may be used:

electron charge: $e = 1.6 \times 10^{-19}$ C

permittivity of free space : $\varepsilon_o = 8.85 \times 10^{-12}$ F/m

relative permittivity of silicon : $\varepsilon_r = 12$

Planck's constant : $h = 6.63 \times 10^{-34}$ J s

Boltzmann's constant : $k = 1.38 \times 10^{-23}$ J/K

speed of light : $c = 3.0 \times 10^{8}$ m/s

The eigenvalue equations for TE modes in a symmetric slab waveguide of thickness d are

$\kappa = k_{1x} \tan(k_{1x} d/2)$ and $\kappa = -k_{1x} \cot(k_{1x} d/2)$

1. You should attempt all parts of this question. Short answers only are required; there is no need to re-state the questions in your answer book, but you should show any calculations you use to arrive at your answers, state any assumptions or approximations made, and give a brief (one or two lines) explanation where appropriate. All parts have equal value. [20]

a) Calculate the relationship between the refractive index and the relative permittivity of a dielectric material.

b) If at some time instant a 2 km length of optical fibre contains 8000 bits of propagating information, what is the bit rate of this signal?

c) Visible light propagates through a sheet of uncoated window glass at approximately normal incidence. If the glass has a refractive index 1.49 in the visible range, estimate the fraction of optical power transmitted.

d) A certain optical receiver detects $2 \cdot 10^{12}$ photons/s, at a nominal wavelength of 1.4 μm. What is the equivalent received optical power in dBm?

e) Calculate the slope efficiency of a laser diode operating at λ_o = 1490 nm if the quantum efficiency is 0.8.

f) Provide an expression for the noise power per unit bandwidth, for thermal noise in an optical detector.

g) If thermal noise and shot noise are equal in a certain optical fibre link, and then the fibre length is decreased by 25%, which of these noise sources will now be greatest at the detector?

h) Explain briefly why silicon is not a favourable semiconductor for constructing optical sources.

i) A silica optical fibre has a refractive index difference Δn of 0.03. Estimate the numerical aperture.

j) Estimate the spectral width in nm of an LED operating at a nominal wavelength of 514 nm.

2. A symmetric slab waveguide with dielectric and cladding refractive indices n_1 and n_2 respectively as shown in Fig. 2.1 has a core thickness d, and a numerical aperture NA =0.21.

a) Find the value d/λ_o, where λ_o is the free space wavelength, for which the m=0 TE mode of this guide has an electric field distribution $E(x)$ such that $E(d/2) = E(0)/\sqrt{2}$. [10]

b) Does the guide that satisfies the condition in (a) support any additional TE modes, and if so how many? [5]

c) If the guide that satisfies the condition in (a) has a core thickness d = 5 μm, at what distance Δx beyond the boundary at $x = d/2$ does the electric field amplitude drop to $\frac{1}{2}E(0)$? [5]

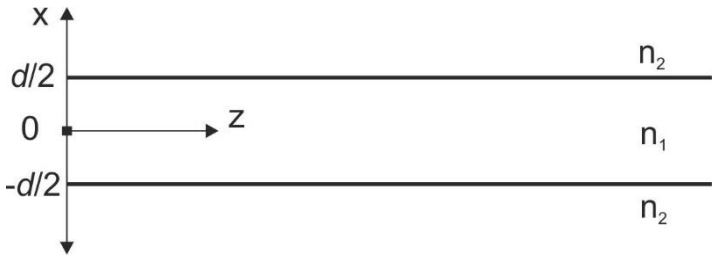


Figure 2.1

3. A laser of peak output power 8 dBm, with nominal wavelength λ_o = 1550 nm and spectral width $\Delta\lambda$ = 0.4 nm, emits into a fibre with an attenuation of 0.2 dB/km and dispersion coefficient 0.75 ps/nm·km. It is modulated with an on-off keying format at a bit rate B. The receiver has quantum efficiency η = 0.8 and input resistance R = 20 kΩ.

a) Assume that the SNR is dominated by thermal noise in the receiver. Hence, derive an expression for the maximum bit rate as a function of fibre length, for an optical SNR of 12. Neglect dispersion, and state any other approximations or assumptions made. [5]

b) Assume that the SNR is dominated by shot noise in the receiver. Hence, derive an expression for the maximum bit rate as a function of fibre length, for an optical SNR of 12. Neglect dispersion, and state any other approximations or assumptions made. [5]

c) Derive an expression for the bit rate at which the dispersion time is equal to one quarter of the bit time $1/B$, as a function of fibre length. [5]

d) On a single graph of bit rate vs. fibre length, sketch the three relations derived in (a), (b) and (c) above, using appropriate scales and ranges for your two axes. Briefly discuss the significance of this graph for the operation of this optical link. [5]

4. A given single mode fibre has a wavelength-dependent effective refractive index n' given by:

$$n' = n_g + \alpha(\lambda_o - \lambda_c)^2 \qquad (4.1)$$

where λ_o is the free-space wavelength and λ_c is the centre operating range wavelength.

a) Calculate expressions for the phase delay τ_p, and the group delay τ_g, for a fibre length L. Note that the group velocity is given by $v_g = d\omega/d\beta$ where β is the propagation constant. [12]

b) Using your solution to (a), derive an expression for the dispersion coefficient D, using $D = \left|(d\tau_g/d\lambda_o)/L\right|$. Hence, show that in this case: $D = \left|\lambda_o(d^2n/d\lambda_o^2)/c\right|$.

Dispersion causes a frequency (or wavelength) dependent phase shift and therefore it can be modelled as an all-pass filter. Derive the expression for a filter function H(ω) equivalent to the dispersion effect of this fibre for a length L. [8]

5. A silicon p-i-n photodiode has p and n doping levels respectively of $N_A = 2\times10^{21}$ m^{-3} and $N_D^+ = 10^{21}$ m^{-3}. The p-layer thickness is 0.6 μm. The attenuation coefficient in Si at the wavelength of interest is given by $\alpha = 0.2\times10^6$ m^{-1}.

a) Find the intrinsic layer thickness w_i such that 85% of photons are absorbed in the intrinsic layer (neglecting Fresnel reflection at the surface). [5]

b) Using w_i as calculated above, find the intrinsic layer doping level N_D^- such that the intrinsic region can be fully depleted by an applied voltage of 4.0 V. [5]

c) Using N_D^- and w_i as calculated above, find the applied bias voltage V such that the electric field amplitude in the intrinsic region varies by 20% (i.e. maximum value is 120% of minimum value). [5]

d) Discuss the main factors to be considered in optimising the electric field magnitude in a p-i-n photodiode. [5]